

## SC2014 : Sensors, Interfacing & Digital Control

### Part 2 : Tutorial 2 – Analogue Control System

**Q1 :** Describe all the possible outcome of a 2<sup>nd</sup> order system (with the aid of diagrams in the s-domain) when the input of the system is a unit step.

**Q2 :** A servomechanism controls the angular position ( $\theta$ ) of a mass through a command signal ( $u$ ). The transfer function in s-domain is given by :

$$\frac{\theta(s)}{u(s)} = \frac{344}{s^2 + 25s + 344}$$

Determine the transient response characteristics (percentage peak overshoot, rise time, settling time, and peak time) of the open loop system?

**Q3 :** The Open-Loop System Model of a 2<sup>nd</sup> Order System is  $\frac{1}{s^2 + 6s + 5}$ . Users find this system to be quite sluggish in the response to a step input. The resident engineer suggest that the response of the system can be improved by creating a unity gain negative feedback control system and adding a simple proportional controller with a gain of  $K$ . Calculate the value of  $K$  if you want the step response to be critically damped ( $\zeta = 1$ )